

We estimate the subgroup's best-fit growth regime by searching over an admissible class $\mathcal{H}$ of candidate CDFs on $[0, 1]$. In this panel, $\mathcal{H}$ is the *Beta regime family*

$$\mathcal{H}_{\text{Beta}} = \{ H_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), 0 < m < 1, \kappa > 0 \},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean SGPC on the 0–1 scale) and κ controls concentration (dispersion of latent SGPC).²

For each candidate $H \in \mathcal{H}$, Panel B provides the induced *predicted* current-grade marginal

$$F_H(v) = \mathbb{E}_U [H(F_0(v \mid U))].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S = \operatorname{argmin}_{H \in \mathcal{H}_{\text{Beta}}} W_1(F_{V,S}^{\text{obs}}, F_H),$$

where

$$W_1(F, G) = \int_0^1 |F(v) - G(v)| \, dv$$

is the Wasserstein-1 (earth mover's) distance between the observed and predicted current-grade marginal CDFs.

The heat map displays $\log_{10}$ Wasserstein-1 distance over $(m, \log_{10} \kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F_{V,S}^{\text{obs}}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., $\text{Beta}(1, 1)$), indicated by the $U(0, 1)$ marker.